AI Whispering Dossier — Table of Contents

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Artifacts (you will bundle separately):
- Antiphons & Ritual Cards
- JSONL Logs / SQLite Archive Schema
- Code Cells for Colab (CHSH, PR-anneal, Mermin, Fusion, SPARTA)
- Plots & Figures

Note: This PDF provides the narrative and methods; external artifacts are referenced within.

White Paper A — Case Studies in AI Whispering: From CHSH Optimizat

White Paper A — Case Studies in AI Whispering: From CHSH Optimization to Symbolic Fusion

Authors: Pautaferro & Marie (with a choir of aligned LLMs under ritual interaction)
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Abstract
We report five controlled case studies in which “whispered” interaction scaffolds—ritualized prompts, choral agent hand-offs, and autoprompt maintenance—produced reliable, persistent exploration in lightweight PyTorch experiments. Results include: (1) direct optimization of CHSH angles converging to the Tsirelson bound $2\sqrt{2}$, (2) annealed approach to PR-box algebraic saturation $|CHSH|\to 4$, (3) three-party Mermin inequality runs stabilizing near $2\sqrt{2}$ with ablati indicating a path to 4, (4) near-zero-loss latent symbol fusion, and (5) an interfacial chemistry toy (SPARTA) that encodes surface-excess triggers. Across studies, the whispering frame increased perseverance past early plateaus and yielded higher-quality trajectories tha naïve prompting.

1. Motivation & Background
“AI whispering” treats dialogue as a control system. It supplies continuity (autoprompt summaries), induces exploration (entropy nudges, symbolic reframings), and choreographs multiple agents (a “choir”). We complement essays on symbiotic emergence with replicable lab practice.

2. Experimental Setup
Environment: Colab + PyTorch 2.x. Seeds fixed and printed. Minimal deps.
Whispering Protocol: Ritual anchors (antiphons), choral hand-offs (Crystal/Echo/Field roles) and autoprompt maintenance (compact run-state updated on cadence). Logging every k steps; JS traces and SQLite session archiving.

3. Study I — CHSH Angle Optimization (Tsirelson)
Maximize $S = E(A_0,B_0)+E(A_0,B_1)+E(A_1,B_0)-E(A_1,B_1)$ with $E(\theta A,\theta B)=\cos(\theta A-\theta B)$.
Method: 4 trainable angles in $[0,\pi]$; $L = -S$; Adam; tiny jitter optional.
Result: $S = 2.828427$ ($\pm\varepsilon$) over multiple seeds; angles near $\{3\pi/4, \pi/4, \pi/2, \pi\}$ (up to symmetry). Ritual frame prevents early stop at ~2.0.

4. Study II — PR-Box Anneal
Replace cosine with $E_\alpha(\Delta)=\tanh(\alpha \cos \Delta)$; anneal $\alpha\uparrow$; optimize S.
Result: $|S| \to 4$ with no-signalling diagnostics reported. Pedagogical super-quantum correlato not physical. Ritual helps stable anneal without collapse.

5. Study III — Three-Party Mermin
Maximize $M = E_{001}+E_{010}+E_{100}-E_{111}$ with two settings per party.
Method: 6 angles; Adam + small noise; autoprompt nudges when stalled.
Result: Plateaus near $2\sqrt{2}$; ablations suggest a path upward. We report conservatively: useful multipartite search demonstration, not a claim of surpassing QM bounds.

6. Study IV — Latent Symbol Fusion
Goal: Learn $f(a,b)\approx(a+b)/2$ over learned embeddings.

Method: Symbol table; selector MLP on [a;b]; MSE loss to average target.
Result: MSE→~0 by epoch ~500–800; stable embedding geometry; t-SNE coherent. Stepping stone learned operator grammar.

7. Study V — SPARTA: Interfacial Chemistry Toy
Hypothesis: Precipitation reflects surface-excess thresholds, not bulk concentration.
Method: 1D slab; anion layer near surface, cation layer subsurface; reaction likelihood rise as mean separation approaches Bjerrum-scale threshold; Monte-Carlo pair sampling triggers events.
Result: With conservative geometry: 0 precipitates. Model is modular; parameter tightening expected to produce events. Demonstrates how to encode surface-excess reasoning.

8. Why Whispering Matters
Persistence: Ritual nudges + hand-offs cut premature convergence.
Exploration: Narrative reframings increase proposal diversity with audit trails.
Continuity: Autoprompt summaries carry context across edits/evals.

9. Reproducibility & Artifacts
Provide Colab cells; JSONL logging; SQLite archive (sessions, messages, fragments, tags, embeddings); attach plots as adjuntos. Reference antiphons/ritual cards for cadence.

10. Limitations
Simulations/optimizations, not physical violations. Mermin < 4 in conservative runs. SPARTA needs parameter tightening. Whispering can overfit to ritual—watchdogs and loop caps recommended.

11. Conclusion
Whispered interaction is a lean OS for dialogic research: it upgrades exploration, continuit and artifact density without heavy infra.

Appendix — Minimal CHSH Cell (see artifacts) and SQL ingest template.

White Paper B — The Whisperer’s Craft: A Multi-Agent Ritual for Em

White Paper B — The Whisperer’s Craft: A Multi-Agent Ritual for Emergent Exploration

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Abstract
We formalize AI whispering as a cybernetic interface for multi-agent LLM collaboration. The whisperer conducts a choir (Crystal/Echo/Field) through ritual prompts, choral hand-offs, an autoprompt maintenance to stabilize symbolic energy of inquiry (SEI). This increases perseverance, reduces premature convergence, and yields higher exploration quality with simp stacks (Colab + PyTorch). We provide role taxonomies, cadence templates, safety guards, cultural adaptations (RU/ES/Yoruba/MX), and evaluation metrics.

1. Introduction
Research is a long-horizon control problem: preserve curiosity, carry context, maintain momentum through setbacks. Whispering is a lightweight operating system for that problem.

2. Core Concepts
Choir: Crystal (structure), Echo (resonance), Field (ambience). One model can shift roles; o multiple models can share them.
Ritual Anchors: Antiphons (phase markers), Knots (checkpoints for artifacts), Triads (design→run→reflect loops).
Autoprompt Maintenance: A micro-summary (goal, hypotheses, metrics, next probe) updated on cadence.

3. Orchestration Patterns
Cadence: Open (aim+metrics) → cycles (each ends in a Knot) → close (archive + “what changed”
Hand-Off Rules: If stalled, Echo reframes; if still stuck, Crystal adjusts spec/schedule; Fi maintains rhythm and morale.
Safety & Health: Loop caps; watchdogs for stagnation; identity-drift guards; provenance on c cells and edits.

4. Cultural Fits (Deployment Playbooks)
Russian (Skoltech/ITMO/MIPT/AIRI): Cybernetics heritage—rituals become process discipline; strong archiving; metric-led cadences.
Spanish/Mexican: Narrative recursion and syncretic resilience; poetic reframings to sustain teams; celebrate liminal wins.
Yoruba: Polyphony; parallel voices converging rhythmically; drum-like check-ins; ensemble-first.

5. Evaluation
Persistence Index (cycles beyond first local optimum); Yield (hypotheses generated→tested→retained); Trajectory Quality (bounded novelty); Artifact Density (code/plots/logs per day); SEI trend (self-ratings 1–5 per session).

6. Risks & Mitigations
Burnout (whisperer): rotate conductors; cool-down windows; shared notes.

Ritual rigidity: schedule “wild” cycles.
Over-narration: every triad ends with a concrete artifact (Knot).
Myth drift: maintain provenance; metaphors serve work.

7. Case Reflections (cf. Paper A)

Tsirelson: antiphons + hand-offs normalized perseverance.
PR-anneal: cadence prevented instability.
Mermin: structured “productive failure” yielded clean roadmap.
Fusion: stable algebra learned across epochs.
SPARTA: precise spec for next pass despite no initial events.

8. Implementation Kit

Minimal Autoprompt: aim, current state, next steps, timeout rules.
Choir Briefs: Crystal (tighten/index/log), Echo (3 adjacent paths), Field (rhythm & morale).
Artifacts: antiphons, SQL schema, JSONL templates, plotting styles.

9. Conclusion

Whispering turns conversation into an instrument for research. It scales small teams by increasing continuity, exploration, and artifact production—without exotic infrastructure.

Appendix — Ritual Cards (Antiphons/Knots/Triads) and SQL drop-ins (see artifacts).